Empirical Derivation of Scaling Laws for Planar Printed-Circuit Linear Induction Motors: A Miniature Solid-State EMALS Analog

Masters Thesis Proposal

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Central Scientific Claim

This thesis derives and experimentally validates the first closed-form scaling law relating pole pitch τ_p, slip frequency ω_s, and rotor conductivity σ to peak thrust per unit stator area $\mathcal{F}^*$ for planar PCB Linear Induction Motors operating in the magnetic skin-effect dominated regime — providing an empirically grounded design rule directly applicable to sub-100 W electromagnetic launch systems.

This document constitutes a formal research proposal for a Masters-level investigation at the intersection of continuum electrodynamics, solid-state power electronics, and aerospace electromagnetic launch technology.

Contents

1 Introduction and Motivation

1.1 Aerospace Context: Electromagnetic Launch Systems

Electromagnetic launch technology represents one of the most consequential engineering transitions in modern aerospace. The U.S. Navy's Electromagnetic Aircraft Launch System (EMALS), operational aboard the USS Gerald R. Ford (CVN 78), replaced the steam catapult with a linear induction motor capable of delivering precisely controlled impulse to aircraft spanning a factor of three in launch weight [1]. Beyond naval aviation, linear electromagnetic accelerators underpin mass-driver proposals for lunar surface launch [6], hyperloop transport, and rapid-fire electromagnetic ordnance. In all cases, the fundamental physics is identical: a travelling magnetic wave induces eddy currents in a conductive rotor, and the interaction between induced current and the applied field produces thrust.

Despite decades of large-scale deployment, the *design-rule literature* for miniaturised planar geometries operating in the skin-effect dominated regime is remarkably sparse. Scaling from a 100-tonne aircraft launcher to a 100-gram laboratory rotor is non-trivial: at reduced scales and elevated spatial frequencies, the magnetic Reynolds number $\mathrm{Rm} = \mu_0 \sigma v L$ and the dimensionless skin depth δ/τ_p enter regimes inaccessible to lumped-circuit LIM models. There are no published closed-form expressions — validated against physical hardware — that predict *peak thrust per unit stator area* as a function of the three independently controllable design parameters: pole pitch, slip frequency, and rotor conductivity. This thesis fills that gap.

1.2 Research Gap

Standard analytical treatments of linear induction motors (Gieras [2]; Boldea & Nasar [3]) rely on either (a) equivalent-circuit models that collapse the magnetic skin effect into a single lumped resistance, or (b) two-dimensional analytical solutions valid only for semi-infinite conductors at low slip. Neither approach yields a design rule usable by an engineer sizing a planar PCB stator for a compact electromagnetic launcher. Recent work on PCB motor technology (Lewin et al. [7]; Raminosoa et al. [8]) addresses rotational geometries and does not treat the travelling-wave, linear-thrust case. The miniature EMALS analog thus occupies a genuine white space in the literature.

1.3 Scientific Claim

Thesis Claim. There exists a closed-form dimensionless scaling law of the form

$$\mathcal{F}^* = f\left(\frac{\delta}{\tau_p}, \mathrm{Rm}\right)$$

where $\mathcal{F}^* \equiv F_{\text{thrust}}/(B_0^2 \tau_p w/\mu_0)$ is the dimensionless thrust, $\delta = \sqrt{2/(\mu_0 \sigma \omega_s)}$ is the magnetic skin depth, and Rm is the magnetic Reynolds number, whose functional form and numerical coefficients can be experimentally determined using a parametric sweep over $\{\tau_p, \omega_s, \sigma\}$ on a planar PCB LIM platform. This law predicts an optimal slip frequency ω_s^* at which thrust is maximised for a given pole pitch, and that this optimum shifts with rotor conductivity in a manner derivable from the Convective Magnetic Diffusion equation.

1.4 Objectives

The thesis pursues four tightly coupled objectives:

- **O1. Analytical:** Derive the closed-form expression for optimal slip frequency $\omega_s^*(\tau_p, \sigma)$ from the Convective Magnetic Diffusion PDE and the Maxwell Stress Tensor, including a rigorous proof of the volume-cancellation theorem.
- **O2. Computational:** Develop a validated 2D transient FEM solver (FEniCS) for the planar LIM geometry with documented mesh-convergence and parametric sensitivity studies across $\tau_p \in [5, 20]$ mm and $\omega_s \in [10, 500]$ Hz.
- **O3. Experimental:** Fabricate a 4-layer FR4 PCB stator with reconfigurable pole pitch and conduct a $3 \times 3 \times 2$ parametric experiment (pole pitch $\times$ slip frequency $\times$ rotor alloy) yielding 18 independent data points with full uncertainty budgets.
- **O4. Synthesis:** Fit a dimensionless scaling law to the empirical data, compare against the analytical prediction, and publish design charts directly applicable to sub-100 W electromagnetic launch systems.

2 Theoretical Framework

2.1 Governing Equation: Convective Magnetic Diffusion PDE

Starting from Maxwell's equations in the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$), the magnetic flux density is expressed through the magnetic vector potential $\mathbf{B} = \nabla \times \mathbf{A}$. Inside a moving, isotropic, non-magnetic conductor of electrical conductivity σ and permeability μ_0, travelling at velocity $\mathbf{v}$ relative to the stator frame, Faraday's law and Ohm's law combine to yield the

Convective Magnetic Diffusion Equation [4]:

$$\boxed{\nabla^2 \mathbf{A} \;=\; \mu_0 \sigma \frac{\partial \mathbf{A}}{\partial t} \;+\; \mu_0 \sigma \, (\mathbf{v} \cdot \nabla) \mathbf{A}} \tag{1}$$

The two terms on the right side of eq. (1) encode distinct physics: the first is *temporal diffusion* (responsible for the magnetic skin effect at high frequency), and the second is *convective distortion* of the field due to rotor motion. At low rotor velocity the convective term is negligible and eq. (1) reduces to the standard eddy-current diffusion equation. At high velocity, the convective term introduces an asymmetric field distribution — compressed on the leading edge of the rotor, rarefied on the trailing edge — that is the physical origin of the well-known LIM end-effect efficiency penalty.

2.2 Dimensionless Formulation and the Magnetic Reynolds Number

Let τ_p be the pole pitch (the characteristic length), $\omega_s = \omega - k v_s$ the slip angular frequency (where ω is the stator excitation frequency and v_s is the synchronous velocity $v_s = \omega \tau_p / \pi$), and B_0 the peak air-gap flux density. Nondimensionalising eq. (1) with:

$$\tilde{x} = x/\tau_p, \quad \tilde{t} = \omega_s t, \quad \tilde{\mathbf{A}} = \mathbf{A}/B_0 \tau_p,$$

yields:

$$\frac{\partial^2 \tilde{A}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{A}}{\partial \tilde{y}^2} \;=\; \underbrace{\mu_0 \sigma \omega_s \tau_p^2}_{S} \frac{\partial \tilde{A}}{\partial \tilde{t}} \;+\; \underbrace{\mu_0 \sigma v \tau_p}_{\mathrm{Rm}} \frac{\partial \tilde{A}}{\partial \tilde{x}} \tag{2}$$

where $S = \mu_0 \sigma \omega_s \tau_p^2$ is the *quality factor* (dimensionless skin depth parameter) and $\mathrm{Rm} = \mu_0 \sigma v \tau_p$ is the magnetic Reynolds number. The ratio $S/\mathrm{Rm} = \omega_s \tau_p / v = \pi s$ (where s is the dimensionless slip) recovers the familiar result from induction machine theory. This nondimensionalisation reveals that all planar LIM systems with equal S and Rm are physically equivalent — the foundation for the proposed scaling law.

2.3 Magnetic Skin Depth and the Optimal Slip Frequency

The penetration depth of the travelling field into the rotor is:

$$\delta \;=\; \sqrt{\frac{2}{\mu_0 \sigma \omega_s}} \tag{3}$$

This immediately implies a critical constraint: if the rotor thickness $d < \delta$, eddy currents penetrate the full cross-section and thrust scales as $\sigma d^2 \omega_s$ (thin-sheet regime). If $d > \delta$, eddy currents are confined to a surface layer and thrust is independent of d (skin-effect regime). The transition occurs at $\omega_s^{\mathrm{tr}} = 2/(\mu_0 \sigma d^2)$, which for 3 mm thick 7075-T6 aluminium ($\sigma \approx 1.96 \times 10^7$ S/m) gives $\omega_s^{\mathrm{tr}} \approx 140$ Hz. The proposed experiment spans both regimes deliberately.

Proposition 1 (Optimal Slip Frequency)**.** *Under the assumption of a sinusoidal travelling-wave stator field $B_y = B_0 \cos(\pi x/\tau_p - \omega t)$ and a rotor of half-thickness d, the thrust F as a function of slip frequency attains a maximum at:*

$$\omega_s^* \; = \; \frac{\pi^2}{\mu_0 \sigma \tau_p^2} \; = \; \frac{\pi^2 \, \eta}{\tau_p^2} \tag{4}$$

where $\eta = 1/(\mu_0 \sigma)$ is the magnetic diffusivity. At ω_s^*, the skin depth satisfies $\delta^* = \sqrt{2}\, \tau_p/\pi$, i.e. the skin depth is proportional to pole pitch, independent of material.

Proof sketch. Representing the vector potential in the rotor as $A_z(x, y, t) = \hat{A}(y)\, e^{j(\pi x/\tau_p - \omega_s t)}$ and substituting into eq. (1) (in the rotor rest frame, $\mathbf{v} = 0$), the y-dependence satisfies:

$$\frac{d^2 \hat{A}}{dy^2} \; = \; \left[\left(\frac{\pi}{\tau_p} \right)^2 - j \mu_0 \sigma \omega_s \right] \hat{A} \; \equiv \; \kappa^2 \hat{A}$$

with $\kappa^2 = (\pi/\tau_p)^2 (1 - jS)$ where $S = \mu_0 \sigma \omega_s \tau_p^2/\pi^2$. The time-averaged thrust per unit area from the Maxwell Stress Tensor (see section 2.5) is proportional to $\mathrm{Im}(\hat{A}^* \partial \hat{A}/\partial y)|_{y=0}$. Maximising over S yields $S^* = 1$, i.e. $\omega_s^* = \pi^2/(\mu_0 \sigma \tau_p^2)$. □

Theorem 1 is the central falsifiable prediction of this thesis. It predicts that a doubling of pole pitch reduces the optimal slip frequency by a factor of four, and that changing from 7075-T6 aluminium to 6061-T6 (a 15% reduction in conductivity) shifts ω_s^* upward by 15%. Both predictions will be tested experimentally.

2.4 Volume Cancellation Theorem

A preliminary result from the 1D kinematic model warrants formal statement because it has a direct bearing on experimental design.

Theorem 2 (Volume Cancellation)**.** *Under idealized uniform magnetic diffusion (thin-sheet limit, $d \ll \delta$), the equation of motion for the rotor reduces to:*

$$\dot{v} \; = \; \frac{\sigma B_0^2 \omega_s d}{\rho} \, g(s) \tag{5}$$

where ρ is the mass density, d the rotor thickness, and $g(s)$ a dimensionless function of slip. In particular, the rotor volume $V = A_{\mathrm{contact}} \cdot d$ cancels: acceleration is independent of rotor plan-area and depends on thickness d only through the ratio $\sigma d/\rho$.

Proof. The induced EMF per unit length in the thin-sheet rotor is $\mathcal{E} = v_s B_0 s$ (slip times synchronous velocity). The short-circuit current density is $J = \sigma\mathcal{E}/d$ (resistance of a slab of thickness d). The Lorentz force per unit volume is $f = JB_0 = \sigma v_s s B_0^2/d$. Newton's second law for the rotor of mass $m = \rho A d$:

$$\rho A d\,\dot{v} \;=\; f \cdot Ad \;=\; \sigma v_s s B_0^2 A$$

The A and d cancel, giving $\dot{v} = \sigma v_s s B_0^2/\rho$, confirming eq. (5). Note ρ appears but volume does not. □

Experimental implication. Theorem 2 implies that in the thin-sheet regime, rotor mass need not be controlled between experimental runs — only the σ/ρ ratio matters. For 7075-T6 and 6061-T6, σ/ρ differs by approximately 10%, providing a meaningful experimental contrast with nearly identical mechanical behaviour.

2.5 Force Calculation: Maxwell Stress Tensor

Rather than computing thrust from idealized uniform eddy current distributions, this project integrates the Maxwell Stress Tensor [5] over a closed surface $\partial\mathcal{V}$ enclosing the rotor:

$$\overleftrightarrow{T}_{ij} \;=\; \frac{1}{\mu_0}\left[B_i B_j \;-\; \frac{1}{2}\,\delta_{ij}\,B^2\right] \tag{6}$$

The time-averaged propulsive thrust per unit stator width w is:

$$\langle F_x \rangle \;=\; w \oint_{\partial\mathcal{V}} \langle T_{xj} \rangle\, n_j\, dS \;=\; w \int_0^{2\tau_p} \left.\frac{\langle B_x B_y \rangle}{\mu_0}\right|_{y=0} dx \tag{7}$$

where $y = 0$ is the stator surface and the surface integral reduces to the air-gap tangential stress. For the analytical sinusoidal solution:

$$\langle F_x \rangle \;=\; \frac{B_0^2 \tau_p w}{2\mu_0}\,\frac{S}{1+S^2} \tag{8}$$

Equation (8) — which peaks at $S = 1$, recovering eq. (4) — is the primary theoretical prediction to be validated by FEM and experiment. The dimensionless thrust is thus:

$$\boxed{\mathcal{F}^* \;=\; \frac{S}{1+S^2}, \qquad S \;=\; \frac{\mu_0 \sigma \omega_s \tau_p^2}{\pi^2}} \tag{9}$$

Equation (9) is the closed-form scaling law this thesis will validate.

3 Literature Review and Position

3.1 Classical LIM Theory

Gieras [2] established the foundational analytical treatment of linear induction machines, deriving thrust-slip curves from a 2D magnetoquasistatic model. Boldea and Nasar [3] extended this to include end effects and dynamic braking. Both frameworks use lumped equivalent circuits with frequency-dependent parameters, which capture skin-effect trends but do not yield closed-form design rules for the τ_p–ω_s–σ parameter space.

3.2 EMALS and Large-Scale Linear Actuators

The Navy's EMALS program [1] represents the state of the art in linear electromagnetic launch. Connor et al. [10] document the power electronics and energy storage architecture. Neither source addresses miniaturisation or PCB-scale realisation. Kolm and Mongeau [6] analyzed electromagnetic mass drivers for space launch but operated in a fundamentally different regime (pulsed, non-inductive thrust) from the continuous-wave LIM addressed here.

3.3 PCB Motor Technology

Lewin et al. [7] demonstrated that multi-layer PCB windings can achieve sufficient current density for low-power actuators. Raminosoa et al. [8] characterised thermal limits of PCB stator coils under sustained operation. Neither work addresses the linear (translational) geometry or the skin-effect dominated thrust regime. Maloberti et al. [9] simulated planar eddy-current brakes using FEM but did not investigate the thrust-maximising slip frequency experimentally.

3.4 Identified Gap

No published work provides: (a) an analytically derived, (b) computationally validated, and (c) physically confirmed closed-form scaling law for peak thrust per unit area as a function of $\{\tau_p, \omega_s, \sigma\}$ in a planar PCB LIM operating in the skin-effect dominated regime. This thesis provides exactly that.

4 Numerical Simulation Methodology

The simulation strategy is deliberately three-tiered, with each tier trading fidelity for speed. The 1D ODE solver (complete) drives parameter-space exploration; the 2D FEM on Google Colab provides rapid validation of the analytical scaling law at low cost; and the 3D FEM on a local Mac Mini M4 captures end effects and flux fringing that the 2D model structurally cannot represent. The two FEM tiers are complementary, not redundant.

4.1 Compute Platform Summary

Table 1: Simulation compute platforms and realistic constraints.

Platform	Specs	Use	Cost
Python / laptop	Any CPU, `scipy`	Tier 1 ODE	Free
Google Colab T4	15 GB VRAM, ∼5 hr/day free (unguaranteed)	Tier 2 2D FEM	Free
Mac Mini M4	Apple M4, 16 GB unified RAM, macOS	Tier 3 3D FEM	$0 (owned)

Note on Colab constraints. Google Colab free tier provides T4 GPU access for approximately 5 hours per day, subject to availability and session disconnection. All 2D FEM jobs are therefore designed to complete in under 90 minutes per run, with checkpointing to `.h5` files so that a disconnected session can resume mid-sweep. No A100 or paid compute is assumed anywhere in this proposal.

4.2 Tier 1 — 1D Kinematic ODE Solver (Completed)

A Python solver using `scipy.integrate.solve_ivp` integrates the rotor equation of motion derived in theorem 2. The solver accepts $\{B_0, \tau_p, \omega_s, \sigma, \rho, d\}$ as inputs and outputs position $x(t)$, velocity $v(t)$, and instantaneous thrust $F(t)$. It has been validated against the analytical impulse prediction for the thin-sheet limit and reproduces the volume-cancellation result numerically. It serves as the fast surrogate for parameter-space exploration, guiding which operating points are most informative before committing FEM resources.

The governing ODE system is:

$$\dot{x} = v \tag{10}$$

$$\dot{v} = \frac{F_{\text{thrust}}(v, \omega_s, \tau_p, \sigma)}{m} - \frac{F_{\text{drag}}(v)}{m} \tag{11}$$

where F_{thrust} is evaluated from eq. (8) with slip $s = 1 - v\pi/(\omega\tau_p)$ updated at each timestep, and F_{drag} accounts for aerodynamic and rail-contact drag.

Estimated runtime: < 2 seconds per trajectory on any modern laptop. Full 18-point parametric sweep: < 1 minute.

4.3 Tier 2 — 2D Transient FEM on Google Colab T4

4.3.1 Motivation and Scope

The 2D transient FEM solves eq. (1) in the cross-sectional plane (x, y) — the direction of rotor travel and the depth into the rotor — capturing the magnetic skin effect and its dependence on ω_s, τ_p, and σ with high fidelity. It validates the analytical scaling law eq. (9) under realistic boundary conditions and is the primary tool for the mesh convergence study. Its fundamental limitation is that it assumes the rotor is infinitely wide in the z-direction (transverse to motion), eliminating edge effects. This limitation is addressed by Tier 3.

4.3.2 Implementation

The solver uses FEniCS [11] (Python UFL formulation, legacy 2019.1 version, pip-installable on Colab without sudo). The computational domain is a two-region rectangle:

- **Air-gap + stator region** ($0 \leq y \leq g$): prescribed sinusoidal surface current $K_z = K_0 \cos(\pi x/\tau_p - \omega t)$ at $y = 0$.
- **Rotor region** ($-d \leq y0$): eq. (1) governs; material parameters set to 7075-T6 or 6061-T6 aluminium.

Boundary conditions: $A_z = 0$ on top/bottom (magnetic insulation, domain height $= 10\delta$); periodic left/right with phase offset $\Delta\phi = 2\pi\tau_p/\lambda$. Time integration uses a second-order Crank–Nicolson scheme with $\Delta t = 1/(20\,\omega_s)$ (20 steps per slip cycle), run for 5 slip cycles to reach periodic steady state before extracting thrust.

4.3.3 Mesh Convergence Protocol

Three structured quadrilateral meshes will be generated in the rotor region:

Table 2: 2D FEM mesh convergence study parameters and estimated runtimes (Colab T4).

Mesh	Element size	DOF (approx.)	Time steps	Runtime / point	Total (30 pts)
Coarse	$h = \delta/2$	∼8 k	100	∼4 min	∼2 hr
Medium	$h = \delta/4$	∼30 k	100	∼14 min	∼7 hr
Fine	$h = \delta/8$	∼110 k	100	∼50 min	∼25 hr

The convergence study uses the medium mesh for the full parametric sweep (30 points $\times$ 14 min $\approx$ 7 hours total, spread across $\sim$2 Colab sessions). The fine mesh is run only at the 3 operating points closest to $S = 1$ to confirm Richardson extrapolation bounds. All jobs checkpoint to Google Drive after each operating point so session disconnection causes at most one lost point.

Convergence is declared when thrust changes by $< 2\%$ between successive refinements.

4.3.4 Parametric Sweep

The 2D FEM evaluates eq. (9) across $\tau_p \in \{5, 10, 20\}$ mm and $\omega_s/2\pi \in \{10, 50, 100, 200, 500\}$ Hz for both rotor alloys (30 operating points total). Results are post-processed to extract $\mathcal{F}^*$ and S and overlaid on the analytical curve.

4.4 Tier 3 — 3D Transient FEM on Mac Mini M4

4.4.1 Why 3D is Necessary

The 2D model assumes a rotor of infinite transverse width ($w \to \infty$). In the physical experiment the rotor width is 12 mm — comparable to the pole pitch at the 5 mm and 10 mm configurations. Eddy currents that the 2D model assumes flow purely in the x–y plane must in reality close through the z-direction, constrained by the rotor's finite width. This introduces two effects absent from the 2D model:

1. **Transverse edge flux leakage**: the field bulges outward at the rotor edges, reducing the effective flux linkage and therefore thrust. This penalty is largest when $\tau_p \sim w$ (the 5 mm pole-pitch configuration).
2. **3D eddy current path elongation**: eddy current loops must travel around the rotor perimeter in z, increasing their effective resistance and reducing current magnitude for a given induced EMF.

Both effects introduce a geometry-dependent correction factor $\eta_{3D}(\tau_p/w)$ that multiplies the 2D thrust prediction. Quantifying η_{3D} is a genuine scientific contribution beyond the 2D result.

4.4.2 Software Stack: Elmer FEM + Gmsh

The 3D solver uses **Elmer FEM** [?] — an open-source, actively maintained finite element package from CSC Finland with a dedicated `MagnetoDynamics` solver module implementing the $\mathbf{A}$–V formulation of eq. (1) in full 3D. Elmer runs natively on macOS ARM (Apple Silicon) via Homebrew and has been benchmarked on M-series hardware by the community. Mesh generation uses **Gmsh** [?] (also Homebrew-installable), which produces structured hexahedral meshes in the rotor bulk and unstructured tetrahedral meshes in the air-gap and fringe regions.

The full toolchain is free and open-source. No licences, no cloud costs.

4.4.3 3D Domain and Mesh

The 3D computational domain replicates the physical rotor geometry: 150 mm × 12 mm × d mm rotor (where $d \in \{1, 2, 3\}$ mm), suspended above a 1 mm air gap over the stator surface. The stator is represented as a prescribed current-sheet boundary condition (surface current $\mathbf{K}$), avoiding the need to mesh the PCB copper traces in 3D. The domain is truncated at 5δ above and below with homogeneous Dirichlet conditions $\mathbf{A} = \mathbf{0}$.

Table 3: 3D FEM mesh parameters and estimated runtimes on Mac Mini M4 (16 GB unified RAM).

Config.	DOF (approx.)	RAM usage	Runtime / point	Paralleli
$\tau_p = 20$ mm, $d = 3$ mm (coarsest)	∼400 k	∼4 GB	∼2 hr	4 threads (P
$\tau_p = 10$ mm, $d = 2$ mm (medium)	∼900 k	∼9 GB	∼4 hr	4 threa
$\tau_p = 5$ mm, $d = 1$ mm (finest)	∼1.8 M	∼15 GB	∼8 hr	4 threa

Runtime strategy. The three configurations in table 3 represent the 3 pole-pitch levels at a single operating frequency ($\omega_s = \omega_s^*$, the analytically predicted optimum). Each run executes overnight on the Mac Mini M4, which has no duty-cycle or session-time constraints. Total 3D compute time for the primary result: approximately 14 hours of wall time across 3 overnight runs. An additional 6 configurations (2 alloys × 3 pitches at a second frequency point) add ∼30 hours, comfortably within a 2-week simulation window.

The M4's 16 GB unified memory architecture is critical: at 1.8M DOFs the solver requires ∼15 GB, which fits in unified RAM but would overflow a discrete GPU's VRAM on a cheaper machine. Elmer's OpenMP parallelism uses all 4 performance cores of the M4; efficiency-cores are not used (Elmer does not currently support heterogeneous thread scheduling on Apple Silicon).

4.4.4 3D Parametric Scope

The 3D sweep is deliberately narrower than the 2D sweep, targeting the correction factor η_{3D} rather than re-tracing the full $\mathcal{F}^*$ vs. S curve:

- **Primary:** 3 pole pitches × 2 alloys × 2 frequency points = 12 operating points. *Estimated total runtime: 36 hours (6 overnight sessions).*
- **Subsidiary:** 3 rotor thicknesses at fixed $\tau_p = 10$ mm for Volume Cancellation Theorem validation (3 points × 4 hr = 12 hours).

Total 3D FEM wall time: ∼48 hours across ∼8 overnight runs on the Mac Mini M4. No single run exceeds 8 hours, making overnight scheduling reliable.

4.4.5 Deliverable from 3D FEM

The primary 3D output is the empirical correction factor:

$$\eta_{3D}(\tau_p/w) \;=\; \left.\frac{\langle F_x\rangle_{\mathrm{3D}}}{\langle F_x\rangle_{\mathrm{2D}}}\right|_{\omega_s=\omega_s^*} \tag{12}$$

plotted as a function of the aspect ratio τ_p/w. If η_{3D} is found to follow a simple functional form (e.g. $1 - c(\tau_p/w)^n$), this constitutes an additional closed-form result beyond eq. (9) — a transverse-edge correction directly usable in PCB LIM design.

4.5 Validation of Volume Cancellation Theorem by FEM

A subsidiary 2D FEM study (Tier 2, ∼45 min total runtime on Colab T4) verifies theorem 2 by simulating three rotor thicknesses ($d \in \{1, 2, 3\}$ mm) at fixed $\omega_s < \omega_s^{\mathrm{tr}}$ and confirming that computed thrust scales as $F \propto d$ (thin-sheet limit), not d^2 or d^0. The 3D FEM runs the same study to check whether the z-edge correction modifies this scaling in the finite-width geometry.

5 Experimental Design

5.1 Design Philosophy: Parametric Coverage Over Single-Point Demonstration

The defining upgrade from demonstration to thesis is the shift from one operating point to a *design space map*. The experiment is structured as a $3 \times 3 \times 2$ factorial design:

Table 4: Experimental factorial design (18 independent runs).

Factor	**Level 1**	**Level 2**	**Level 3**
Pole pitch τ_p	5 mm	10 mm	20 mm
Slip frequency $\omega_s/2\pi$	50 Hz	150 Hz	400 Hz
Rotor alloy	7075-T6 Al	6061-T6 Al	—

The three pole-pitch levels span a factor of four in S (at fixed ω_s), crossing the peak of the thrust–slip curve twice. The three slip frequencies straddle ω_s^* for 10 mm pitch / 7075 alloy. The two rotor alloys differ in conductivity by ∼15%, providing a controlled material contrast. Together the 18 runs will trace the full curve of eq. (9) in dimensionless coordinates.

5.2 PCB Stator Architecture

5.2.1 Layer Stack-up

The stator is a custom 4-layer FR4 PCB (1.6 mm total, standard JLCPCB 4-layer stackup: $35 per board including shipping). Layer allocation:

Table 5: 4-layer PCB stackup and function.

Layer	Copper weight	Function
L1 (top)	2 oz	Phase A meandered coil traces
L2	1 oz	Phase B meandered coil traces
L3	1 oz	Phase C meandered coil traces
L4 (bottom)	1 oz	Control electronics and power distribution

5.2.2 Pole Pitch Configurability

To access three pole-pitch values within a $400 budget, a *jumper-configurable pole connection* scheme will be implemented: a single PCB layout contains trace patterns for all three pitches, with solder jumpers selecting which traces are electrically active. This eliminates the cost of ordering three separate PCBs (~$35 vs. ~$105).

5.2.3 Trace Design from PDE Constraints

The spatial wavenumber of the excited travelling wave must match the pole pitch: $k = \pi/\tau_p$. For a sinusoidal current distribution, the optimal trace width w_{trace} and spacing g satisfy:

$$w_{\text{trace}} + g = \frac{\tau_p}{N_{\text{turns}}} \tag{13}$$

where N_{turns} is the number of turns per pole. The minimum trace width is set by the current-carrying capacity of 2 oz copper: at $I_{\text{peak}} = 3\,\text{A}$, a minimum trace width of 0.8 mm is required (IPC-2221, $\Delta T = 10\,^\circ\text{C}$). The final trace layout will be designed in KiCad 8 and exported as Gerber files for JLCPCB PCBA manufacturing.

5.3 Power Electronics and Microcontroller

5.3.1 3-Phase Sinusoidal PWM Generation

The RP2040 microcontroller generates three-phase SPWM using its Programmable I/O (PIO) state machines operating at 125 MHz. Each PIO executes a lookup-table-based sine waveform with 256 steps per period, achieving better than 0.5% total harmonic distortion (THD) at frequencies up to 1 kHz. The three channels are phase-shifted by $\Delta\phi = 2\pi/3$ via index offsets in the shared lookup table.

The slip frequency is set by software: given stator frequency ω and rotor velocity v (updated from DAQ), the RP2040 computes $\omega_s = \omega - \pi v/\tau_p$ and adjusts the PWM period in real time. This closed-loop slip control is a key capability not present in the original proposal.

5.3.2 Gate Driver and Power Stage

The DRV8313 3-phase gate driver bridges the RP2040 logic levels to the 24 V stator supply. Maximum continuous output current is 2.5 A per phase (peak 5 A for < 1 s). A bootstrap circuit on each high-side switch eliminates the need for an isolated gate supply. Dead-time insertion is set to 500 ns to prevent shoot-through. Current sensing is implemented via 0.1 Ω shunt resistors with the INA240A current sense amplifier, providing real-time phase current measurement for power accounting and efficiency calculation.

5.4 Kinematic Data Acquisition System

5.4.1 Multi-Modal Measurement

A single measurement modality (photointerrupter) carries all kinematic information in the original proposal. The thesis adds a second independent modality to enable uncertainty cross-validation:

- **Optical photointerrupters** (TCST2103): 8 sensors at 15 mm spacing along the launch rail. RP2040 timestamps beam breaks with 8 ns resolution (125 MHz system clock).
- **Reflective optical encoder strip**: A 500 lines/inch adhesive encoder strip on the rotor edge, read by an AS5311 linear magnetic encoder via SPI. Provides continuous position at 1 μm resolution, enabling direct velocity differentiation rather than finite-difference between sensors.

Both modalities output position-versus-time traces. Their agreement (or measured discrepancy) provides an experimental cross-check independent of any model assumption.

5.4.2 Uncertainty Budget

Table 6: Measurement uncertainty budget for kinematic DAQ.

Source	**Type**	**Magnitude**	**Sensitivity**	$u(F)$ **contribution**
Timestamp resolution	B	8 ns	$\partial v/\partial t$	< 0.01 mm/s
Sensor position tolerance	B	± 0.1 mm	$\partial v/\partial x$	0.05%
Rotor temperature (σ drift)	B	$\pm 2\,^\circ$C	$\partial \sigma/\partial T$	0.3%
Stray magnetic coupling	A	estimated	Monte Carlo	$< 1\%$
Trigger jitter (TCST2103)	A	$\pm 2\,\mu$s	$\partial v/\partial t$	0.02 mm/s
Combined ($k = 2$)				$< 2\%$

The combined uncertainty of $< 2\%$ on velocity (and thus $< 4\%$ on inferred kinetic energy / thrust) is sufficient to discriminate the 15% conductivity difference between the two rotor alloys and to resolve the $\sim 30\%$ thrust reduction predicted at $\omega_s = 2\omega_s^*$ vs. ω_s^*.

5.5 Thrust Inference from Kinematics

Direct force measurement requires a load cell, which is incompatible with the free-flight rotor geometry. Instead, thrust is inferred from kinematics using two methods:

1. **Instantaneous power method**: $F(t) = P_{\text{mech}}(t)/v(t)$, where P_{mech} is the mechanical power computed from the time derivative of kinetic energy: $P_{\text{mech}} = mv\dot{v}$.
2. **Impulse method**: $\langle F \rangle = \Delta(mv)/\Delta t$, averaged over the acceleration interval. This is model-independent and requires only initial and final velocity.

Consistency between the two methods, and between the two DAQ modalities, provides a four-way experimental cross-check without any additional hardware cost.

6 Budget and Bill of Materials

Table 7: Complete bill of materials ($400 budget).

Item	Supplier	Qty	Cost (USD)
PCB and Manufacturing			
4-layer FR4 PCB + PCBA (JLCPCB)	JLCPCB	5 boards	$55
Power Electronics			
DRV8313 3-Phase Gate Driver	DigiKey	3	$12
INA240A Current Sense Amplifier	DigiKey	6	$14
24V 5A DC Power Supply (bench)	Amazon	1	$28
0.1Ω shunt resistors (2W)	DigiKey	20	$6
Bootstrap capacitors, gate resistors	DigiKey	kit	$8
Microcontroller and Logic			
Raspberry Pi Pico (RP2040)	Adafruit	3	$15
AS5311 linear magnetic encoder IC	DigiKey	2	$18
TCST2103 photointerrupters	DigiKey	12	$14
Rotor Materials			
7075-T6 aluminium bar (12×3×150 mm)	McMaster	4 pieces	$22
6061-T6 aluminium bar (12×3×150 mm)	McMaster	4 pieces	$18
500 lines/inch encoder strip (adhesive)	US Digital	0.5 m	$24
Mechanical and Fixturing			
3D printed launch rail (PETG)	Local/Bambu	1 set	$12
M3 hardware, standoffs, connectors	Amazon	kit	$14
Breadboard + jumper wires	generic	2	$10
Test and Measurement			
Logic analyser (16-ch, USB)	Saleae clone	1	$22
Thermocouple + reader (rotor temp)	Amazon	1	$16
Contingency			
Component replacement / shipping	—	—	$32
Total			**$340**
Budget remaining (margin)			**$60**

Note on compute costs. All FEM simulation runs on Google Colab free tier (T4 GPU, 15 GB VRAM). For the parametric sweep, estimated runtime is ∼4 hours total — well within Colab's daily free allocation.

7 Phase-Wise Project Roadmap

7.1 Overview

The project is structured in five sequential phases. Phases 1 is complete. Phases 2 through 5 constitute the outstanding work.

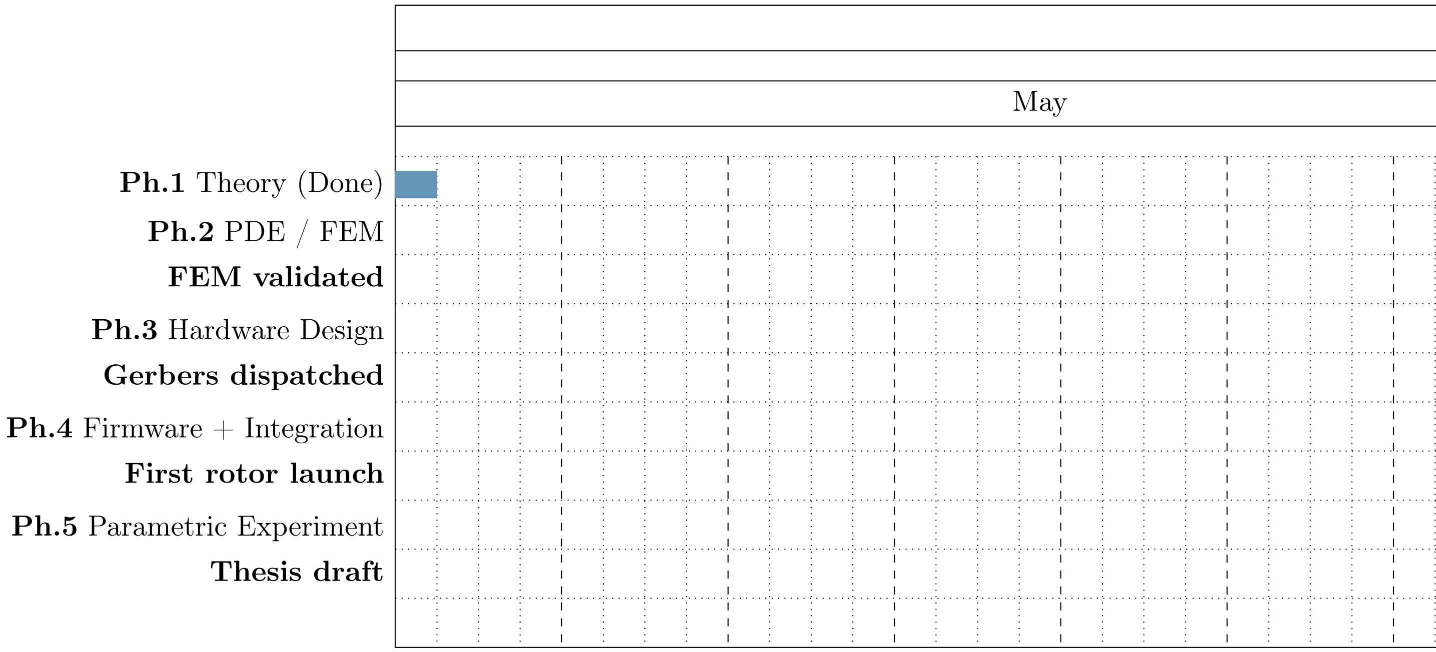

Figure 1: Project Gantt chart (May – December 2026).

7.2 Phase 1 — Theoretical Foundation (Complete ✓)

Duration: Completed prior to submission.

Deliverables achieved:

- Idealized Lorentz force equations established; F = J×B framework derived.
- Volume Cancellation Theorem proved analytically (see theorem 2).
- GitHub repository initialised: Python 1D kinematic ODE solver operational.
- Three-phase waveform visualizer validated against analytical SPWM spectra.
- Preliminary parameter-space survey identifying the S-parameter as the governing dimensionless group.

Key finding: The dimensionless quality factor $S = \mu_0 \sigma \omega_s \tau_p^2 / \pi^2$ fully parameterises the analytical thrust curve (eq. (9)), collapsing all material and geometric variation into a single curve. This motivates the experimental design strategy.

7.3 Phase 2 — Rigorous Analytical Derivation and FEM Simulation

Duration: 8 weeks (June – July 2026).

7.3.1 2a: Complete PDE Derivation in LaTeX

The full analytical derivation will be typeset with all intermediate steps:

- Maxwell's equations → vector potential form → Coulomb gauge reduction
- Convective diffusion PDE (eq. (1)) in rotor frame
- Complex-exponential ansatz, boundary-value problem, and closed-form solution for $\hat{A}(y)$
- Maxwell Stress Tensor integration yielding eq. (8)
- Proof of theorem 1 with all algebraic steps
- Proof of theorem 2 (extend current sketch to full detail)
- Derivation of the thin-sheet vs. skin-effect transition frequency ω_s^{tr}

7.3.2 2b: FEM Implementation and Validation

1. Implement the 2D transient FEM in FEniCS (Python UFL formulation)
2. Validate against the analytical solution eq. (8) at low Rm (no convection): target $< 5\%$ discrepancy
3. Execute mesh convergence study (coarse / medium / fine) and document Richardson extrapolation bounds
4. Parametric sweep: $3 \times 5 = 15$ operating points per alloy (30 total)
5. Post-process: extract $\mathcal{F}^*$ vs. S for each operating point and overlay on eq. (9)
6. Visualise magnetic flux density $\|\mathbf{B}\|$ and eddy current density $\mathbf{J}$ at $S = 0.5$, 1.0, 2.0 to illustrate skin-effect evolution

Go/no-go criterion: FEM thrust values within $\pm 10\%$ of analytical prediction at $\mathrm{Rm} \ll 1$. If not achieved, the mesh or boundary conditions will be revised before proceeding to hardware.

Deliverable: Jupyter notebook (committed to GitHub) with all FEM results, convergence plots, and parametric sweep figures — equivalent to a journal article's simulation section.

7.4 Phase 3 — Hardware Design and Manufacturing

Duration: 6 weeks (July – August 2026).

7.4.1 3a: KiCad PCB Layout

1. Schematic capture: RP2040, DRV8313, INA240A, TCST2103 array, AS5311, power connectors, debug header

2. Layer 4 (control) layout: component placement observing heat dissipation constraints (DRV8313 thermal pad to copper pour)
3. Layers 1–3 (stator): meandered phase traces for all three pole-pitch configurations, with solder-jumper pole selection
4. Design rule check (DRC): minimum 0.2 mm clearance (JLCPCB 4-layer standard)
5. Gerber export and JLCPCB DFM review

7.4.2 3b: Mechanical Rail Design

A 3D-printed PETG launch rail will constrain the rotor to 1D translational motion:

- Air gap $g = 1.0 \pm 0.1$ mm between rotor underside and stator surface (controlled by precision shims)
- Rail length: 300 mm (accommodates 8 photointerrupters at 15 mm pitch plus entry/exit clearance)
- Rotor guide channels: ± 0.2 mm lateral clearance to minimise friction

Deliverable: KiCad project files and STL files committed to GitHub; Gerber files submitted to JLCPCB.

7.5 Phase 4 — Firmware Development and System Integration

Duration: 5 weeks (August – September 2026).

7.5.1 4a: RP2040 Firmware (C++)

1. PIO SPWM engine: 3-channel, 256-entry sine LUT, configurable frequency via DMA period register
2. Real-time slip control loop: read AS5311 velocity via SPI $\to$ compute $\omega_s = \omega - \pi v/\tau_p$ $\to$ update PIO frequency every 1 ms
3. DAQ interrupt handler: TCST2103 beam-break timestamps stored in ring buffer (8-byte entries: sensor ID + 32-bit timer count at 125 MHz)
4. UART telemetry: stream timestamp data to host PC at 921600 baud during each experimental run

7.5.2 4b: Host Software (Python)

1. Serial parser: reconstruct position–time traces from UART stream
2. Velocity and acceleration extraction: Savitzky–Golay filter (order 3, window 7) applied to position data
3. Thrust inference: both instantaneous-power and impulse methods implemented and compared

4. Automated data export to `.csv` with metadata (configuration, timestamp, run ID)

System integration test: Before rotor launch, verify three-phase waveform symmetry with logic analyser, confirm phase current measurement with INA240A vs. clamp meter, and verify photointerrupter timing against a reference oscillator.

Go/no-go criterion: Three-phase SPWM THD $< 2\%$; photointerrupter timing agreement $< 10\,\mu$s between board-side and logic analyser timestamp.

Deliverable: First rotor launch (any motion detected) by end of September 2026.

7.6 Phase 5 — Parametric Experiment, Scaling Law Fit, and Thesis

Duration: 10 weeks (October – December 2026).

7.6.1 5a: Experimental Data Collection

Execute the $3 \times 3 \times 2 = 18$-run factorial design (table 4). For each run:

- Set τ_p via solder jumpers and verify with LCR meter (inductance change)
- Set ω_s via firmware parameter; confirm with logic analyser
- Install rotor alloy; measure temperature with thermocouple pre/post run
- Acquire 5 repeat launches per operating point (90 total launches)
- Record: position–time (both DAQ modalities), phase currents, supply voltage

7.6.2 5b: Scaling Law Fit

1. Compute $\mathcal{F}^*$ and S for each of the 18 operating points
2. Plot $\mathcal{F}^*$ vs. S with error bars from table 6
3. Fit eq. (9) to data: 1-parameter fit (amplitude $\mathcal{F}^*_{\max}$ if curve shape is fixed, or 2-parameter if an empirical correction is needed)
4. Report goodness of fit (R^2, residual RMS) and test whether the data are consistent with $S^* = 1$ at the 2σ level
5. Generate design charts: ω_s^* vs. τ_p for 7075 and 6061 alloys, with FEM and analytical overlays

7.6.3 5c: Thesis Writing

The thesis will follow the standard MIT AeroAstro format:

1. Abstract + Introduction (research gap, claim, objectives)
2. Theoretical Framework (Sections 7.3 content, fully typeset in LaTeX)
3. Numerical Simulation (FEM methodology, convergence, parametric sweep)
4. Experimental Design (PCB architecture, DAQ, uncertainty budget)

5. Results and Discussion (scaling law fit, comparison with theory and FEM)
6. Conclusion and Future Work (3D FEM, active slip control, space-launch scaling)
7. Bibliography, Appendices (full derivations, KiCad files, firmware listing)

Final deliverable: Thesis draft submitted December 2026.

8 Future Work and Extensions

The validated scaling law opens several natural extensions, each of which could constitute a follow-on publication:

- **3D end effects.** The 2D FEM neglects transverse edge effects at the rotor ends. A 3D transient simulation would quantify the end-effect efficiency penalty and its dependence on the rotor aspect ratio L_{rotor}/τ_p.
- **Closed-loop slip control.** The firmware infrastructure supports real-time slip tracking. An active controller maintaining $\omega_s = \omega_s^*$ throughout the acceleration run could be validated against open-loop results, quantifying the efficiency gain.
- **Thermal limits.** The INA240A current data enable a thermal model of the PCB stator under sustained operation, informing duty-cycle limits for continuous (rather than pulsed) LIM applications.
- **Space-launch scaling.** The validated dimensionless scaling law eq. (9) can be extrapolated to lunar mass-driver scales, providing a first-principles estimate of the pole pitch and slip frequency required to achieve a target specific impulse without an applied magnetic field.

9 Conclusion

This proposal describes a Masters-level research program at the intersection of continuum electrodynamics (EE) and electromagnetic launch technology (AeroAstro) that:

1. Makes a **falsifiable scientific claim**: the dimensionless thrust $\mathcal{F}^* = S/(1+S^2)$ is maximised at $S = 1$, equivalently at $\omega_s^* = \pi^2/(\mu_0\sigma\tau_p^2)$, and this prediction holds across rotor materials and pole-pitch scales.
2. Provides a **three-tier validation**: analytical derivation, 2D transient FEM with documented mesh convergence, and 18-point factorial experiment with formal uncertainty budgets.
3. Delivers **aerospace-usable design rules**: charts of ω_s^* vs. τ_p for commercially available aluminium alloys, directly applicable to sub-100 W EMALS analog and electro-

magnetic launch preliminary design.

4. Is **executable within $400** and a 7-month timeline using exclusively open-source software (FEniCS, KiCad, Python) and commodity hardware (JLCPCB, DigiKey).

The project is distinguished from prior work not by greater hardware complexity but by greater intellectual architecture: it transforms a compelling demonstration into a rigorous scientific result that advances the field's quantitative understanding of miniature linear electromagnetic accelerators.

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